Branch Coverage Report (Mini-Project Part-3 §2.b.iv)

Course: Software Engineering | **Section:** A | **Team Number:** 9
Target System: Online Exam Portal (OEP)
Testing Window: 2 – 6 November 2026 (Sprint 3); results regenerated 12 November 2026 after the v1.0.1 fixes
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Code tested: commit `57e01fd` on `main` · **Repository:** https://github.com/vinnieee7/SE-Online-Exam-Portal-Team09

1. Summary

Branch coverage counts every outcome of a decision — both sides of each `if` / `else`, each `?:` result, each `switch` case and each short-circuit of `&&`, `||` and default values — and reports how many were taken at least once.

Item	Value
Branch outcomes in the backend	413
Taken by the tests	247
Branch coverage	**59.80%**
Tests run	85 (all suites), all passing
Tool / run	Jest 30.5.2 + Istanbul, 12 November 2026, commit `57e01fd`

2. Branches by file

File	Branches	Taken	Not taken	Branch %
`src/controllers/examController.js`	150	79	71	52.66
`src/controllers/evalController.js`	84	54	30	64.28
`src/controllers/deliveryController.js`	65	35	30	53.84
`src/controllers/authController.js`	47	29	18	61.70
`src/app.js`	16	10	6	62.50
`src/middleware/auth.js`	16	15	1	93.75
`src/config/db.js`	13	8	5	61.53
`src/middleware/logger.js`	12	8	4	66.66
`src/db/seed.js`	10	9	1	90.00
All files	**413**	**247**	**166**	**59.80**

Route files and `secureServer.js` contain no branches (they only wire handlers) and are 100% covered by statements.

3. Decisions where both outcomes are tested

3.1 Authentication middleware (`auth.js`, 15 of 16 outcomes)

```
Request ──► token present? ──no──► 401                          (TC-Sec-03a)
                 │yes
                 ▼
            signature valid? ──no──► 403                        (TC-Sec-03b, 03c, 03d)
                 │yes
                 ▼
            account locked? ──yes──► 403                         (TC-Auth-02b, TC-Sec-03f)
                 │no
                 ▼
            role allowed? ──no──► 403                            (TC-Auth-04, TC-Sec-03e, UT-Auth-03)
                 │yes
                 ▼
              next()
```

The one outcome not taken: a valid token for a user who has since been deleted (401).

3.2 Other decisions with both sides tested

Decision	True side	False side
Exam end time after start time (`createExam`)	Exam created — `TC-Exam-01`	400 — `TC-Exam-03`, DEF-004
Linked question ids exist (`createExam`)	Exam created — `TC-Exam-01`	400, not 500 — `TC-Exam-01b`
Attempt already exists (`startOrGetAttempt`)	Resume, same order — `TC-Take-05b`	New attempt — `TC-Take-01`, `TC-Take-05a`
Time left > 0 (`Math.max(0, …)`)	Seconds left — `TC-Take-02a`, `02b`	0 — `TC-Take-02d`
Window end before full duration	Capped at window end — `TC-Take-02c`	Full duration — `TC-Take-02a`
Timestamp already has a time zone	Parsed as is	Treated as UTC — `TC-Take-02e`, DEF-002
Question type objective (`autoGrade`)	Graded at once — `TC-Eval-01`	Left for the instructor — `TC-Eval-02`
Scorecard owner is the requester	Returned — `TC-Eval-03`	403 — `TC-Sec-06`, DEF-005
Email already registered (case-insensitive)	409 — `TC-Auth-01a`, DEF-003	201 — `TC-Auth-01`
Password matches	Token issued — `TC-Auth-02`	401 — `TC-Auth-02a`, `TC-Sec-07a`
Request body is valid JSON with required fields	Handled	400 with no internal detail — `TC-Sec-07e`, DEF-012

4. Branches not taken

File	Not taken	What they are
`src/controllers/examController.js`	71	`updateExam` (every field-by-field change and the publish switch), `deleteExam`, `deleteQuestion`, and missing-field checks in `createExam` / `addQuestion`
`src/controllers/deliveryController.js`	30	Exam not found / not published / not started / already ended; reopening a submitted attempt; invalid auto-save bodies; the unreachable pre-merge grading fallback
`src/controllers/evalController.js`	30	Invalid manual-grade input (empty grade list, unknown attempt, attempt still in progress), scorecard before solutions are released, `getMyScorecard`, analytics for an unknown exam
`src/controllers/authController.js`	18	Missing registration fields, reset request for an unknown email, wrong or expired reset code, `updateUserStatus`
`src/app.js`	6	Fallback stub routers for modules not yet merged, and the `PORT` default
`src/config/db.js`	5	Database open / query error callbacks
`src/middleware/logger.js`	4	IP fallbacks when neither proxy header nor socket address exists
`src/middleware/auth.js`, `src/db/seed.js`	2	Deleted-user token; seed run as a script rather than imported

Most untaken branches are input-validation and error-handling paths. The planned next tests, in order: (1) update and delete an exam and a question, (2) Admin role change and lock through the API, (3) starting an exam before it opens and after it closes, (4) invalid manual-grade input.

5. Conclusion

Branch coverage is **59.80%** (247 of 413 outcomes). Every security decision (token, signature, lock, role, ownership) and every decision tied to a logged defect has both outcomes tested.

Prepared by Aadish Sarin (QA Lead).